

Q5. The following pseudocode tries to implement selection sort on a list [64, 25, 12, 22, 11] to sort in descending order. It contains THREE errors. Identify them and rewrite the correct pseudocode.

```
BEGIN
SET list = [64,25,12,22,11]
SET n = 5
FOR i = 1 TO n
SET max_pos = i
FOR j = i + 1 TO n
IF list[j] > list[max_pos] THEN
max_pos = j
ENDIF
ENDFOR
SWAP list[i] AND list[max_pos]
ENDFOR
OUTPUT list
END
```

Identified Errors

1. **Incorrect Starting Index (0-based vs 1-based):** Python lists are 0-indexed. The outer loop starts at 1 (FOR i = 1 TO n), which means it completely skips checking the very first element (index 0) of the array.
2. **Index Out of Bounds:** The loops iterate up to n (which is 5). In a 5-element, 0-indexed list, the valid indices are only 0 through 4. Accessing list[5] will cause the program to crash with an "Index Out of Bounds" error.
3. **Inefficient Outer Loop Bound:** The outer loop doesn't need to run all the way to the end of the list. By the time you reach the last element, the rest of the list is already sorted, so it will naturally be in its correct place. The outer loop should stop at n-2 (the second-to-last element). *(Note: Additionally, line 9 is missing the SET keyword used everywhere else in the original snippet).*

Corrected Pseudocode

```
BEGIN
SET list = [64, 25, 12, 22, 11]
SET n = 5
FOR i = 1 TO n - 1
  SET max_pos = i
  FOR j = i + 1 TO n
    IF list[j] > list[max_pos] THEN
      SET max_pos = j
    ENDIF
  ENDFOR
  SWAP list[i], list[max_pos]
ENDFOR
OUTPUT list
END
```